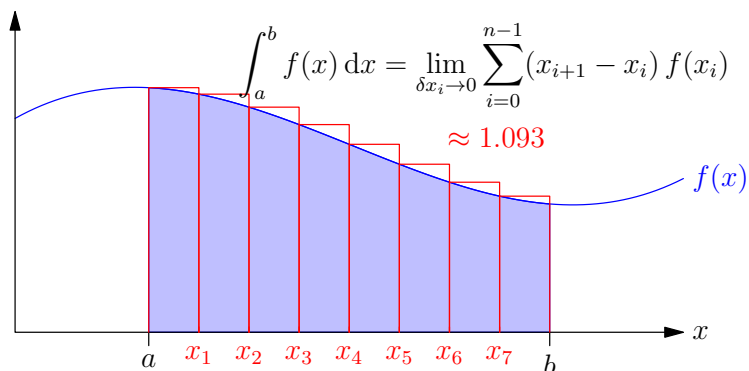
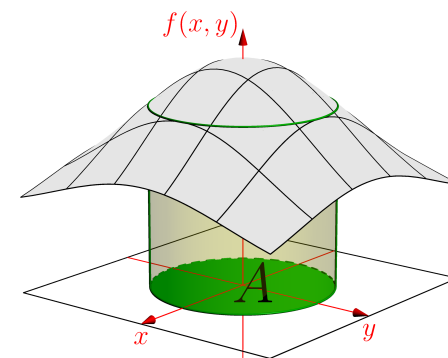


## Integral Calculus



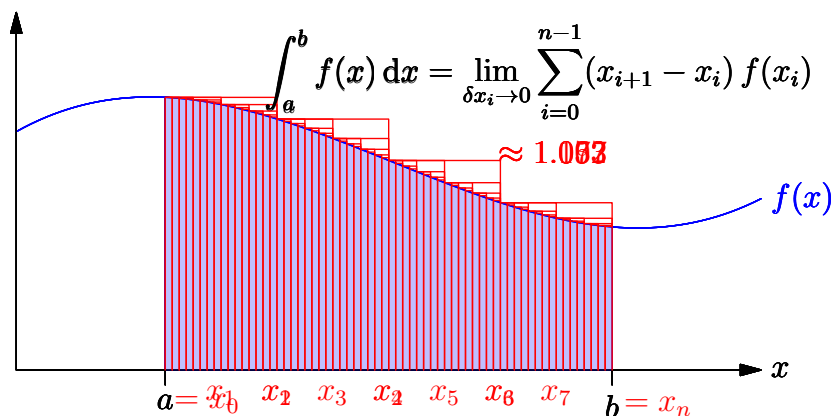
*Riemann Integration, integration by parts, gaussian integrals*

1. Defining Integrals
2. Doing Integrals
3. Gaussian Integrals



## Riemann Integral

- Integrals represent area beneath a curve



## Linearity of Integration

- Integration is a linear operator

$$\begin{aligned}
 \int_a^b (r f(x) + s g(x)) dx &= \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) (r f(x_i) + s g(x_i)) \\
 &= \lim_{\delta x_i \rightarrow 0} \left( \sum_{i=0}^{n-1} (x_{i+1} - x_i) r f(x_i) + \sum_{i=0}^{n-1} (x_{i+1} - x_i) s g(x_i) \right) \\
 &= \lim_{\delta x_i \rightarrow 0} \left( r \sum_{i=0}^{n-1} (x_{i+1} - x_i) f(x_i) + s \sum_{i=0}^{n-1} (x_{i+1} - x_i) g(x_i) \right) \\
 &= r \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) f(x_i) + s \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) g(x_i) \\
 &= r \int_a^b f(x) dx + s \int_a^b g(x) dx
 \end{aligned}$$

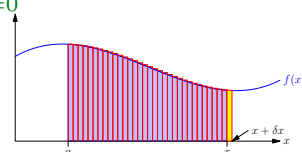
## Fundamental Law of Calculus

- Let

$$I(a, x) = \int_a^x f(z) dz = \lim_{\delta z_i \rightarrow 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i)$$

- Now for small  $\delta x$

$$I(a, x + \delta x) = \int_a^{x+\delta x} f(z) dz = \lim_{\delta z_i \rightarrow 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i) + \delta x f(x)$$



- Thus

$$\frac{dI(a, x)}{dx} = \lim_{\delta x \rightarrow 0} \frac{I(x + \delta x) - I(x)}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{\delta x f(x)}{\delta x} = f(x)$$

## The Other Way Around

- Consider

$$\begin{aligned} \int_a^b \frac{df(x)}{dx} dx &= \int_a^b \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} dx \\ &= \lim_{x_{i+1} - x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \\ &= \lim_{x_{i+1} - x_i \rightarrow 0} \sum_{i=0}^{n-1} (f(x_{i+1}) - f(x_i)) \\ &= (f(x_1) - f(x_0)) + (f(x_2) - f(x_1)) + (f(x_3) - f(x_2)) + \dots \\ &\quad + (f(x_{n-1}) - f(x_{n-2})) + (f(x_n) - f(x_{n-1})) \\ &= f(x_n) - f(x_0) = f(b) - f(a) \end{aligned}$$

- We can think of integration as an **anti-derivative**—it undoes differentiation

## Indefinite Integrals

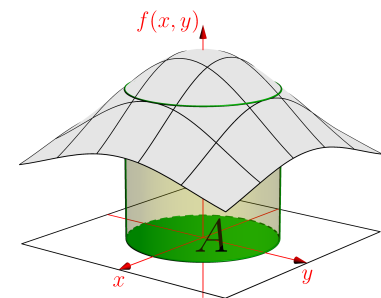
- So far we have considered **definite integrals** where we integrate between two points ( $a$  and  $b$ )
- However, when think about integration as an anti-derivative, it is useful to think of a function  $F(x) = \int f(x) dx$
- So that  $F'(x) = f(x)$
- However the function  $F(x)$ ,  $F(x) + 1$ ,  $F(x) + \pi$ , etc. all have the same derivative so  $F(x)$  is only defined up to an additive constant
- Note that the definite integral is given by

$$\int_a^b f(x) dx = F(b) - F(a)$$

## Multiple Integrals

- For functions involving many independent variables (e.g.  $f(x, y)$ ,  $f(x, y, z)$ ,  $f(x)$ ) we can integrate over multiple dimensions
- For example

$$\iint_A f(x, y) dx dy$$

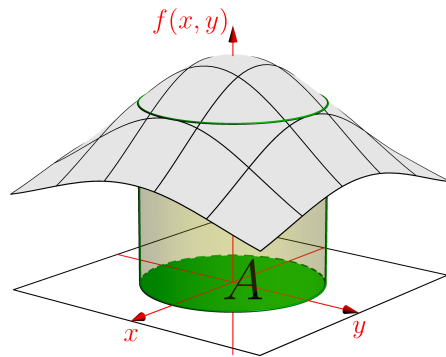


- It gets tedious writing multiple integral signs and I tend to write just one

$$\int \dots \int f(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n = \int f(x) dx$$

## Outline

1. Defining Integrals
2. **Doing Integrals**
3. Gaussian Integrals



## Performing Integration

- A key method for performing integrals is through knowledge of the anti-derivative
- If we know  $F'(x) = f(x)$  then  $F(x) + c = \int f(x) dx$
- E.g. we know that  $dx^n/dx = nx^{n-1}$  therefore

$$\int x^{n-1} dx = \frac{1}{n} \int \frac{dx^n}{dx} dx = \frac{x^n}{n} + c$$

and

$$\int_a^b x^{n-1} dx = \frac{b^n}{n} - \frac{a^n}{n}$$

## Is Integration Straightforward?

- We saw, due to the product and chain rules, that we can differentiate almost anything. Given integration is the anti-derivative can we integrate anything?
- Products and compositions

$$\int f(x)g(x) dx = ? \quad \int f(g(x)) dx = ?$$

- Unfortunately, unlike differentiation we don't have a small parameter we can expand in
- In general integration is hard

## Integration by Parts

- Recall the product rule  $\frac{d}{dx} f(x)g(x) = \frac{d}{dx} f(x)g(x) + f(x)\frac{d}{dx} g(x)$
- Integrating we get

$$\begin{aligned} \int_a^b \frac{d}{dx} f(x)g(x) dx &= \int_a^b \frac{d}{dx} f(x)g(x) dx + \int_a^b f(x)\frac{d}{dx} g(x) dx \\ &= [f(x)g(x)]_a^b = f(b)g(b) - f(a)g(a) \end{aligned}$$

- Unfortunately we get two integrals, but we can turn this around

$$\int_a^b f(x)\frac{d}{dx} g(x) dx = [f(x)g(x)]_a^b - \int_a^b \frac{d}{dx} f(x)g(x) dx$$

whether this is helpful depends on  $f(x)$  and  $g(x)$

## Example of Integration by Parts

- Consider

$$\begin{aligned}\Pi(z) &= \int_0^\infty x^z e^{-x} dx = \int_0^\infty x^z \frac{d(-e^{-x})}{dx} dx \\ &= [x^z (-e^{-x})]_0^\infty - \int_0^\infty \frac{dx^z}{dx} (-e^{-x}) dx \\ &= \int_0^\infty (zx^{z-1}) e^{-x} dx = z \int_0^\infty x^{z-1} e^{-x} dx = z \Pi(z-1)\end{aligned}$$

- Thus  $\Pi(z) = z \Pi(z-1)$ , but

$$\Pi(0) = \int_0^\infty e^{-z} dz = [-e^{-x}]_0^\infty = -e^{-\infty} - (-e^0) = 1$$

- Now

$$\Pi(n) = n \Pi(n-1) = n(n-1) \Pi(n-2) = n(n-1)(n-2) \dots 1 = n!$$

## Substitution

- We can make a transformation from  $x$  to  $u = u(x)$

$$\begin{aligned}\int_a^b f(x) dx &= \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} f(x_i) (x_{i+1} - x_i) \\ &= \lim_{\delta u_i \rightarrow 0} \sum_{i=0}^{n-1} f(x(u_i)) \frac{x(u_{i+1}) - x(u_i)}{u_{i+1} - u_i} (u_{i+1} - u_i) \\ &= \int_{u(a)}^{u(b)} f(x(u)) \frac{dx(u)}{du} du\end{aligned}$$

- where  $u_i$  is such that  $x(u_i) = x_i$  or  $u_i = u(x_i)$  where  $u(x)$  is the inverse of  $x(u)$

- using  $\lim_{\delta u_i \rightarrow 0} \frac{x(u_{i+1}) - x(u_i)}{u_{i+1} - u_i} = \frac{dx(u_i)}{du}$

## Example of Integration by Substitution

- We consider  $I(n) = \int_0^\infty x^n e^{-x^2/2} dx$
- Let  $u(x) = x^2/2$  or  $x(u) = \sqrt{2u}$  so that

$$\frac{dx(u)}{du} = \frac{1}{\sqrt{2u}} \quad u(0) = 0 \quad u(\infty) = \infty$$

- Thus

$$\begin{aligned}I(n) &= \int_0^\infty (\sqrt{2u})^n e^{-u} \frac{1}{\sqrt{2u}} du \\ &= 2^{\frac{n-1}{2}} \int_0^\infty u^{\frac{n-1}{2}} e^{-u} du = 2^{\frac{n-1}{2}} \Pi\left(\frac{n-1}{2}\right)\end{aligned}$$

- $I(1) = 1$ ,  $I(3) = 2 \times 1! = 2$ ,  $I(5) = 2^2 \times 2! = 8$  but  $I(0) = \Pi(-1/2)/\sqrt{2}$ ,  $I(2) = \sqrt{2} \Pi(1/2) = \Pi(-1/2)/\sqrt{2}$

## Changing Variables in Multidimensional Space

- When changing variables in many dimensions  $\mathbf{x} \rightarrow \mathbf{u}$  the change of variables involves the Jacobian

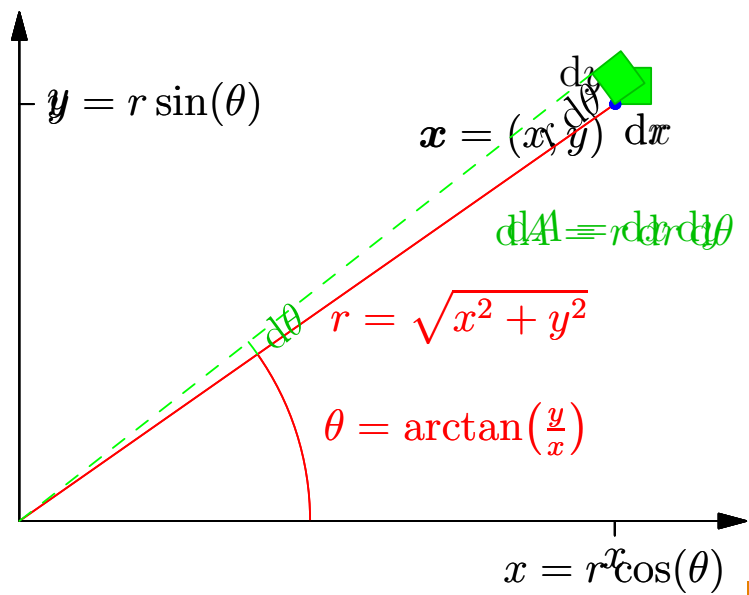
$$\int f(\mathbf{x}) d\mathbf{x} = \int f(\mathbf{x}(\mathbf{u})) |\det(\mathbf{J})| d\mathbf{u}, \quad \mathbf{J} = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \dots & \frac{\partial x_1}{\partial u_n} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \dots & \frac{\partial x_2}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \frac{\partial x_n}{\partial u_2} & \dots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}$$

- E.g. transforming from Cartesian coordinates  $(x, y)$  to polar coordinates  $(r, \theta)$  then  $x = r \cos(\theta)$  and  $y = r \sin(\theta)$

$$\begin{aligned}|\det(\mathbf{J})| &= \left| \det \begin{pmatrix} \frac{\partial r \cos(\theta)}{\partial r} & \frac{\partial r \cos(\theta)}{\partial \theta} \\ \frac{\partial r \sin(\theta)}{\partial r} & \frac{\partial r \sin(\theta)}{\partial \theta} \end{pmatrix} \right| = \left| \det \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix} \right| \\ &= r (\cos^2(\theta) + \sin^2(\theta)) = r\end{aligned}$$

- That is,  $dx dy = r dr d\theta$

## Change of Variables in Pictures



## Differentiating Through the Integral

- A trick that sometimes works is differentiating through an integral, e.g. consider finding moments

$$M_n = \mathbb{E}[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx$$

- We can define a momentum generating function

$$Z(\ell) = \int_{-\infty}^{\infty} e^{\ell x} f_X(x) dx$$

- Then  $M_n = Z^{(n)}(0)$  (the  $n^{\text{th}}$  derivative of  $Z(\ell)$  at  $\ell = 0$ )

$$\left. \frac{d^n Z(\ell)}{d\ell^n} \right|_{\ell=0} = \int_{-\infty}^{\infty} \left. \frac{d^n e^{\ell x}}{d\ell^n} \right|_{\ell=0} f_X(x) dx = \int_{-\infty}^{\infty} x^n f_X(x) dx = M_n$$

## Cumulant Generating Function

- Note that  $e^{\ell x} = 1 + \ell x + \frac{1}{2}\ell^2 x^2 + \frac{1}{3!}\ell^3 x^3 + \dots$

- So

$$Z(\ell) = \int_{-\infty}^{\infty} e^{\ell x} f_X(x) dx = 1 + \ell M_1 + \frac{1}{2}\ell^2 M_2 + \frac{1}{3!}\ell^3 M_3 + \dots$$

- Now using  $\log(1 + \epsilon) = \epsilon - \frac{1}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 + \dots$

$$G(\ell) = \log(Z(\ell)) = \ell M_1 + \frac{1}{2}\ell^2 (M_2 - M_1^2) + \frac{1}{3!}\ell^3 (M_3 - 3M_2 M_1 + 2M_1^3) + \dots$$

- So that  $\kappa_n = G^{(n)}(0)$ , with  $\kappa_1 = M_1$  (the mean),  $\kappa_2 = M_2 - M_1^2$  (the variance),  $\kappa_3 = M_3 - 3M_2 M_1 + 2M_1^3$  (the third cumulant related to the skewness)

## More Integration

- Although we have a few tricks, integration is hard
- Surprisingly integration sometimes is easier when carried out in the complex plane
- This is a beautiful part of mathematics (due largely to Cauchy)—but beyond the scope of this course
- Interestingly, also there is an algorithm that allows us to integrate a lot of function. It is sufficiently complicated that you need to write a computer algorithm of considerable complexity to implement it. Most symbolic manipulation packages (e.g. Mathematica) have implemented some part of this algorithm

## Special Functions

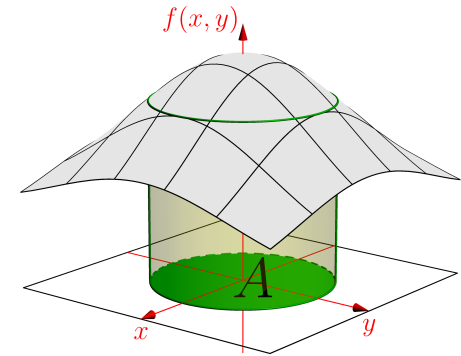
- There are integrals with no known closed form solution■
- We saw that  $\Pi(z) = \int_0^\infty x^z e^{-x} dx$  satisfies  $\Pi(z) = z\Pi(z-1)$ ■
- For integer  $n$  then  $\Pi(n) = n!$ ■ but for general  $z$ , the integral  $\Pi(z)$  can't be written in terms of elementary functions■
- We consider  $\Pi(z)$  as a special function in its own right■
- Although, history has left us with the gamma function instead

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx = \Pi(z-1)$$

- Other special function defined by integrals exist (e.g. the Bessel , Aire, hypergeometric, elliptic, error functions, . . . )■

## Outline

1. Defining Integrals
2. Doing Integrals
3. **Gaussian Integrals**



## Gaussian Integrals

- Gaussian integrals are integrals involving  $e^{-x^2}$ , e.g.

$$\int_{-\infty}^{\infty} e^{-x^2} dx \quad \int_{-\infty}^{\infty} x^4 e^{-ax^2-bx} dx$$

- They are important in computing integrals with respect to the normal distribution

$$\mathcal{N}(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

- The great news is that these integrals are all doable■
- The bad news is that they are quite tricky to do■

## The Gaussian Integral

- The integral over a Gaussian is surprisingly difficult

$$I_1 = \int_{-\infty}^{\infty} e^{-x^2/2} dx$$

- There is a nice trick which is to consider

$$I_1^2 = \int_{-\infty}^{\infty} e^{-x^2/2} dx \int_{-\infty}^{\infty} e^{-y^2/2} dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy$$

- Making the change of variables  $r = \sqrt{x^2 + y^2}$  and  $\theta = \arctan(y/x)$  (so that  $x = r \cos(\theta)$ ,  $y = r \sin(\theta)$  and  $x^2 + y^2 = r^2$ )

$$I_1^2 = \int_0^{2\pi} d\theta \int_0^\infty r e^{-r^2/2} dr = 2\pi \int_0^\infty r e^{-r^2/2} dr$$

## The Gaussian Integral Continued

- From before

$$I_1^2 = 2\pi \int_0^\infty r e^{-r^2/2} dr$$

- Finally let  $u = r^2/2$  so that  $du/dr = r$  or  $du = r dr$  we get

$$I_1^2 = 2\pi \int_0^\infty e^{-u} du = 2\pi$$

- So that  $I_1 = \sqrt{2\pi}$
- Incidentally,  $I_1 = \sqrt{2}\Pi(-1/2)$  so  $\Pi(-1/2) = \Gamma(1/2) = \sqrt{\pi}$

## Multi-dimensional Gaussians

- Consider

$$I_3 = \int_{-\infty}^\infty \cdots \int_{-\infty}^\infty e^{-\frac{1}{2}\|\mathbf{x}\|_2^2} dx_1 \cdots dx_n$$

where  $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$

- Note that  $\|\mathbf{x}\|_2^2 = x_1^2 + x_2^2 + \cdots + x_n^2$  and using  $e^{\sum_i a_i} = \prod_i e^{a_i}$

$$\begin{aligned} I_3 &= \int_{-\infty}^\infty \cdots \int_{-\infty}^\infty e^{-\frac{1}{2} \sum_{i=1}^n x_i^2} dx_1 \cdots dx_n \\ &= \prod_{i=1}^n \int_{-\infty}^\infty e^{-x_i^2/2} dx_i = \prod_{i=1}^n \sqrt{2\pi} = (2\pi)^{n/2} \end{aligned}$$

## Normal Distribution

- We consider

$$I_2 = \int_{-\infty}^\infty e^{-(x-\mu)^2/(2\sigma^2)} dx$$

- Making the change of variables  $z = (x - \mu)/\sigma$  so that  $dz = dx/\sigma$  or  $dx = \sigma dz$ . Then

$$I_2 = \sigma \int_{-\infty}^\infty e^{-z^2/2} dz = \sigma I_1 = \sqrt{2\pi} \sigma$$

- Note that the *probability density function* (PDF) for a normally distributed random variable is given by

$$\mathcal{N}(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

## Full Multi-variate Normal

- Consider

$$I_4 = \int_{-\infty}^\infty \cdots \int_{-\infty}^\infty e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Xi}^{-1}(\mathbf{x}-\boldsymbol{\mu})} dx_1 \cdots dx_n$$

- Let  $\boldsymbol{\Xi}^{-1} = \mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^T$  and make the change of variables  $\mathbf{y} = \mathbf{V}^T(\mathbf{x} - \boldsymbol{\mu})$
- The Jacobian  $\mathbf{J}$  has elements (note that  $\mathbf{x} = \mathbf{V}\mathbf{y} + \boldsymbol{\mu}$ )

$$J_{ij} = \frac{\partial x_i}{\partial y_j} = \frac{\partial}{\partial y_j} \left( \sum_{k=1}^n V_{ik} y_k + \mu_i \right) = V_{ij}$$

- So that  $\mathbf{J} = \mathbf{V}$  and consequently  $|\det(\mathbf{J})| = |\det(\mathbf{V})| = 1$  then

$$I_4 = \int_{-\infty}^\infty \cdots \int_{-\infty}^\infty e^{-\frac{1}{2}\mathbf{y}^T \boldsymbol{\Lambda}^{-1}\mathbf{y}} dy_1 \cdots dy_n = \prod_{i=1}^n \int_{-\infty}^\infty e^{-y_i^2/(2\lambda_i)} dy_i = \prod_i \sqrt{2\pi\lambda_i}$$

## Determinants

- Using the facts, that  $\Xi = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$  then

$$\det(\Xi) = \det(\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T) = \det(\mathbf{V})\det(\mathbf{\Lambda})\det(\mathbf{V}^T) = \det(\mathbf{\Lambda}) = \prod_{i=1}^n \lambda_i$$

using  $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$  and  $\det(\mathbf{V}) = 1$

- Recall  $I_4 = \prod_i \sqrt{2\pi\lambda_i} = (2\pi)^{n/2} \sqrt{\det(\Xi)}$
- We note for an  $n \times n$  matrix  $\mathbf{M}$  then  $\det(c\mathbf{M}) = c^n \det(\mathbf{M})$  so that

$$I_4 = (2\pi)^{n/2} \sqrt{\det(\Xi)} = \sqrt{\det(2\pi\Xi)}$$

- Finally, we get that for the PDF of a normal to integrate to 1

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \Xi) = \frac{1}{\sqrt{\det(2\pi\Xi)}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Xi^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

■

## Summary

- Integration is extra-ordinarily useful as a tool of analysis
- It occurs when you work with probabilities densities for continuous random variables
- Integration is beautiful, but hard—often impossible
- Normal distributions lucky almost always give rise to integrals that can be computed in closed form, although often it requires quite a bit of work
- Making friends with integration will give you a super-power that not too many people share